Customer Shopping Behavior Analysis

3,900 purchases — 18 columns — goal: uncover spending, segments, preferences, subscriptions



Made with GAMMA

Dataset Summary

Scope

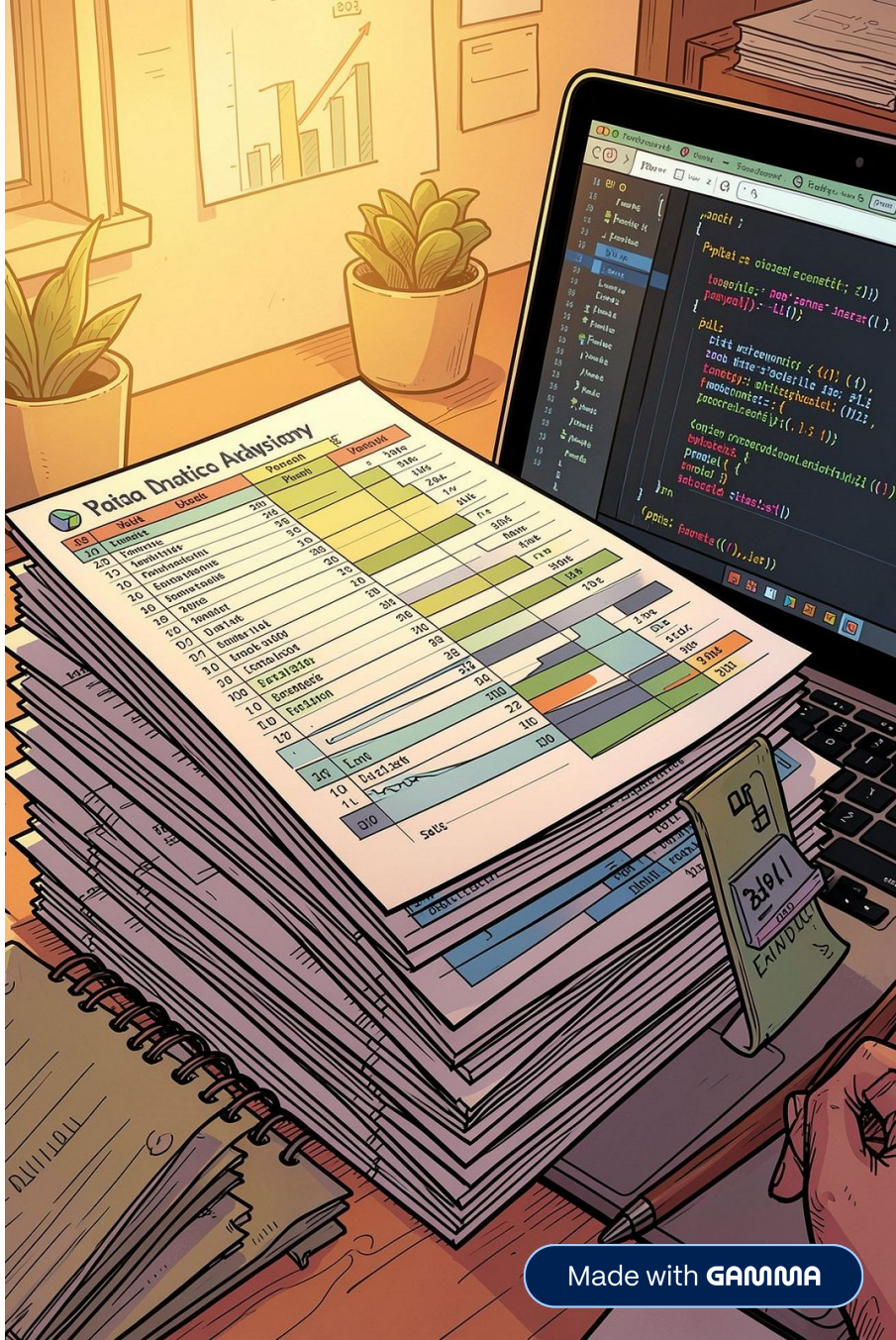
Rows: 3,900 · Columns: 18

Key Features

Demographics, purchase details, behavior, shipping, discounts

Missing Data

37 nulls in Review Rating





Data Preparation (Python)

01

Load & Explore

pandas, df.info(), .describe()

02

Missing & Impute

Imputed Review Rating by category median

03

Standardize

Renamed columns to snake_case

04

Engineer

age_group, purchase_frequency_days

Made with GAMMA



SQL Analysis (MySQL)

DB Integration

Cleaned dataframe loaded to MySQL database

Structured Queries

Revenue, segments, product and shipping analyses

Made with GAMMA

Revenue & Subscriptions

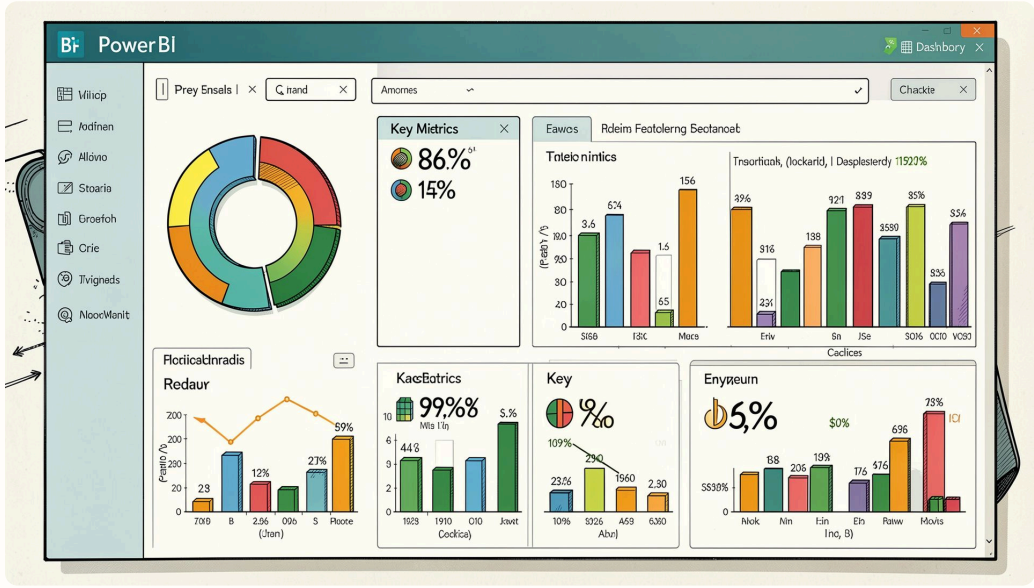
Revenue by Gender

Female: 75,191 · Male: 157,890

Subscribers

Yes: 1,053 (total revenue 62,645) · No: 2,847 (total revenue 170,436)

Dashboard Snapshot



Made with GAMMA

Product Insights



Top by Rating

Gloves 3.86 · Sandals 3.84 · Boots 3.82 · Hat 3.80 · Skirt 3.78



Top Sellers

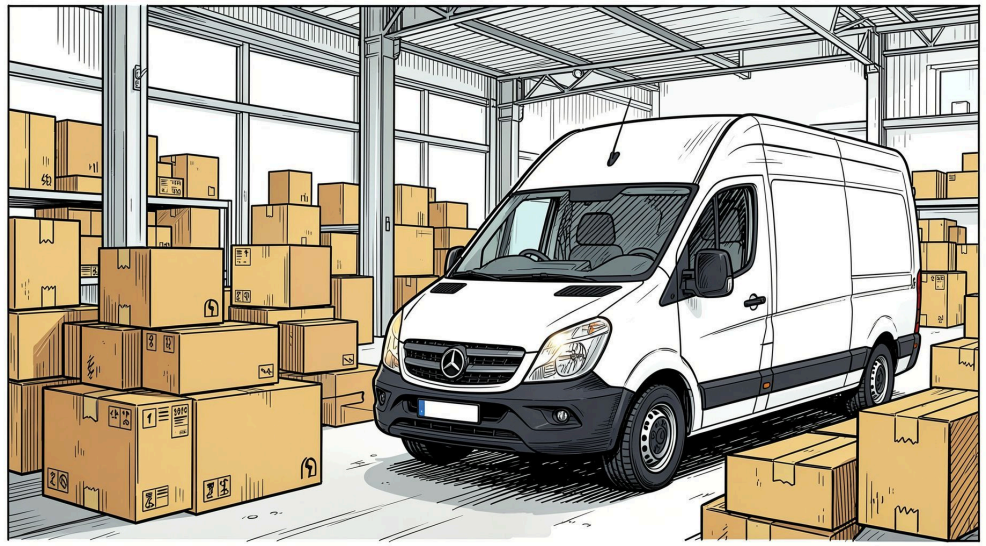
Clothing: Blouse, Pants, Shirt · Accessories: Jewelry, Sunglasses, Belt



Discount-Dependent

Hat 50% · Sneakers 49.66% · Coat 49.07% · Sweater 48.17% · Pants 47.37%

Made with GAMMA



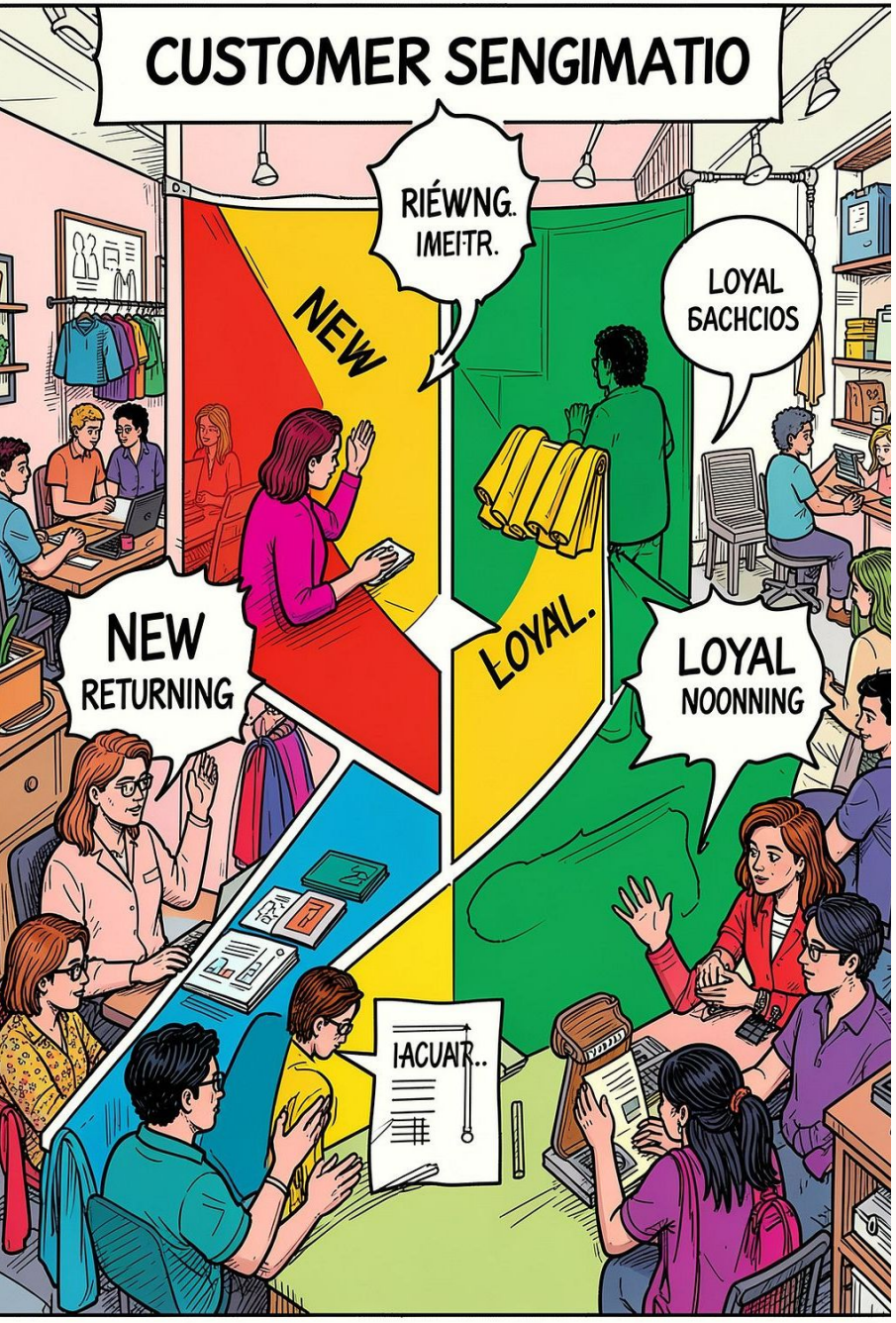
Shipping & High-Spend Discounts

Avg purchase: Standard $58.46 · Express $60.48

High-spending discount users identified — 839 rows examined

Sample high spenders: customer_ids with purchase_amounts: 64,73,90,85,...

Made with GAMMA



Customer Segments

Loyal

3,116 customers

Returning

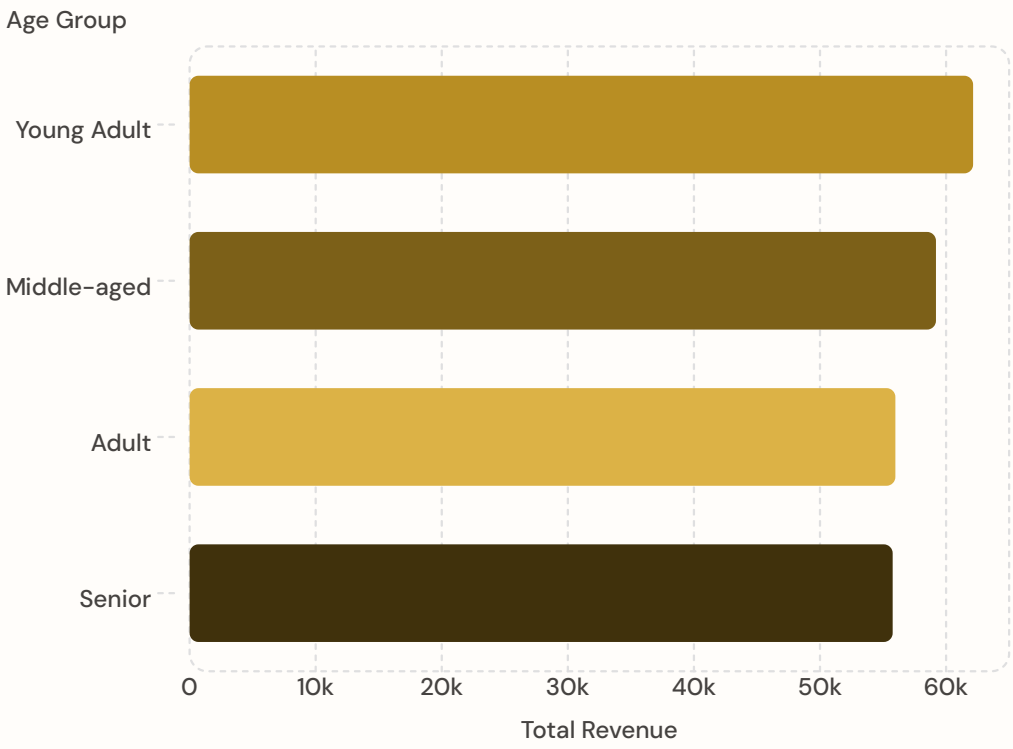
701 customers

New

83 customers

Repeat buyers (>5 purchases): Yes 958 · No 2,518

Made with GAMMA



Revenue by Age Group

Young Adult leads (~62K) · Middle-aged, Adult, Senior ~55–59K

Action: target high-revenue age groups for campaigns

Made with GAMMA

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to increase loyalty

Discount Policy

Balance promotions with margin control

Product Positioning

Feature top-rated and best-selling items

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

Made with GAMMA